

SIMATS ENGINEERING

Department of Computer Science & Engineering

CO5-ClassActivity

Programming in Java - File I/O Operations

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Course Code	CSA0904
Course Name	Programming in Java
Assignment Topic	CO5-ClassActivity: File Deletion and Recreation
Question No.	Question 5

QUESTION 5: DELETE FILE IF EXISTS AND RECREATE

Problem Statement:

Create a Java program to check if a specified file (sample_file.txt) exists. If it exists, delete it successfully, recreate a fresh file instance with the same name, and write initial data into it.

Source Code (RecreateFile.java):

```
import java.io.File;
import java.io.FileWriter;
import java.io.IOException;

public class RecreateFile {
    public static void main(String[] args) {
        String fileName = "sample_file.txt";
        File file = new File(fileName);

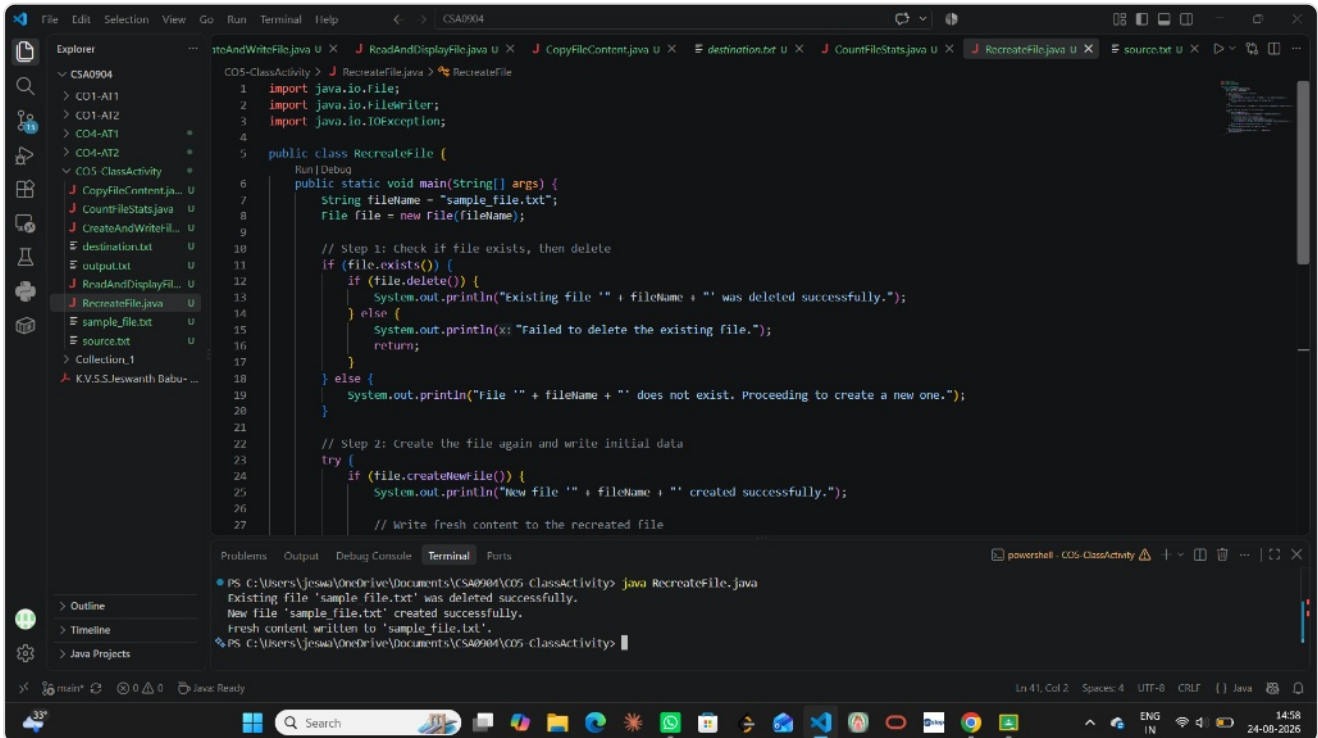
        // Step 1: Check if file exists, then delete
        if (file.exists()) {
            if (file.delete()) {
                System.out.println("Existing file '" + fileName + "' was deleted successful");
            } else {
                System.out.println("Failed to delete the existing file.");
                return;
            }
        } else {
            System.out.println("File '" + fileName + "' does not exist. Proceeding to create");
        }

        // Step 2: Create the file again and write initial data
        try {
            if (file.createNewFile()) {
                System.out.println("New file '" + fileName + "' created successfully.");

                // Write fresh content to the recreated file
                try (FileWriter writer = new FileWriter(file)) {
                    writer.write("This is a newly created fresh file.\n");
                    writer.write("Recreated after deleting the previous version successfully.");
                }
                System.out.println("Fresh content written to '" + fileName + "'.");
            } else {
                System.out.println("Could not create the file.");
            }
        } catch (IOException e) {
            System.err.println("File operation error: " + e.getMessage());
            e.printStackTrace();
        }
    }
}
```

EXECUTION & VERIFICATION OUTPUT

Program Execution & Terminal Output:



The screenshot displays an IDE with a project named 'CSA0904'. The Explorer panel on the left shows the project structure, including files like 'RecreateFile.java'. The main editor shows the code for 'RecreateFile.java', which implements a logic to delete an existing file and create a new one with fresh content. The code is as follows:

```
1 import java.io.File;
2 import java.io.FileWriter;
3 import java.io.IOException;
4
5 public class RecreateFile {
6     public static void main(String[] args) {
7         String fileName = "sample_file.txt";
8         File file = new File(fileName);
9
10        // Step 1: Check if file exists, then delete
11        if (file.exists()) {
12            if (file.delete()) {
13                System.out.println("Existing file '" + fileName + "' was deleted successfully.");
14            } else {
15                System.out.println("Failed to delete the existing file.");
16                return;
17            }
18        } else {
19            System.out.println("File '" + fileName + "' does not exist. Proceeding to create a new one.");
20        }
21
22        // Step 2: Create the file again and write initial data
23        try {
24            if (file.createNewFile()) {
25                System.out.println("New file '" + fileName + "' created successfully.");
26            }
27            // Write fresh content to the recreated file
```

The Terminal panel at the bottom shows the execution output for the command `java RecreateFile.java`:

```
PS C:\Users\jesaa\OneDrive\Documents\CSA0904\CO5 ClassActivity> java RecreateFile.java
Existing file 'sample_file.txt' was deleted successfully.
New file 'sample_file.txt' created successfully.
Fresh content written to 'sample_file.txt'.
PS C:\Users\jesaa\OneDrive\Documents\CSA0904\CO5 ClassActivity>
```